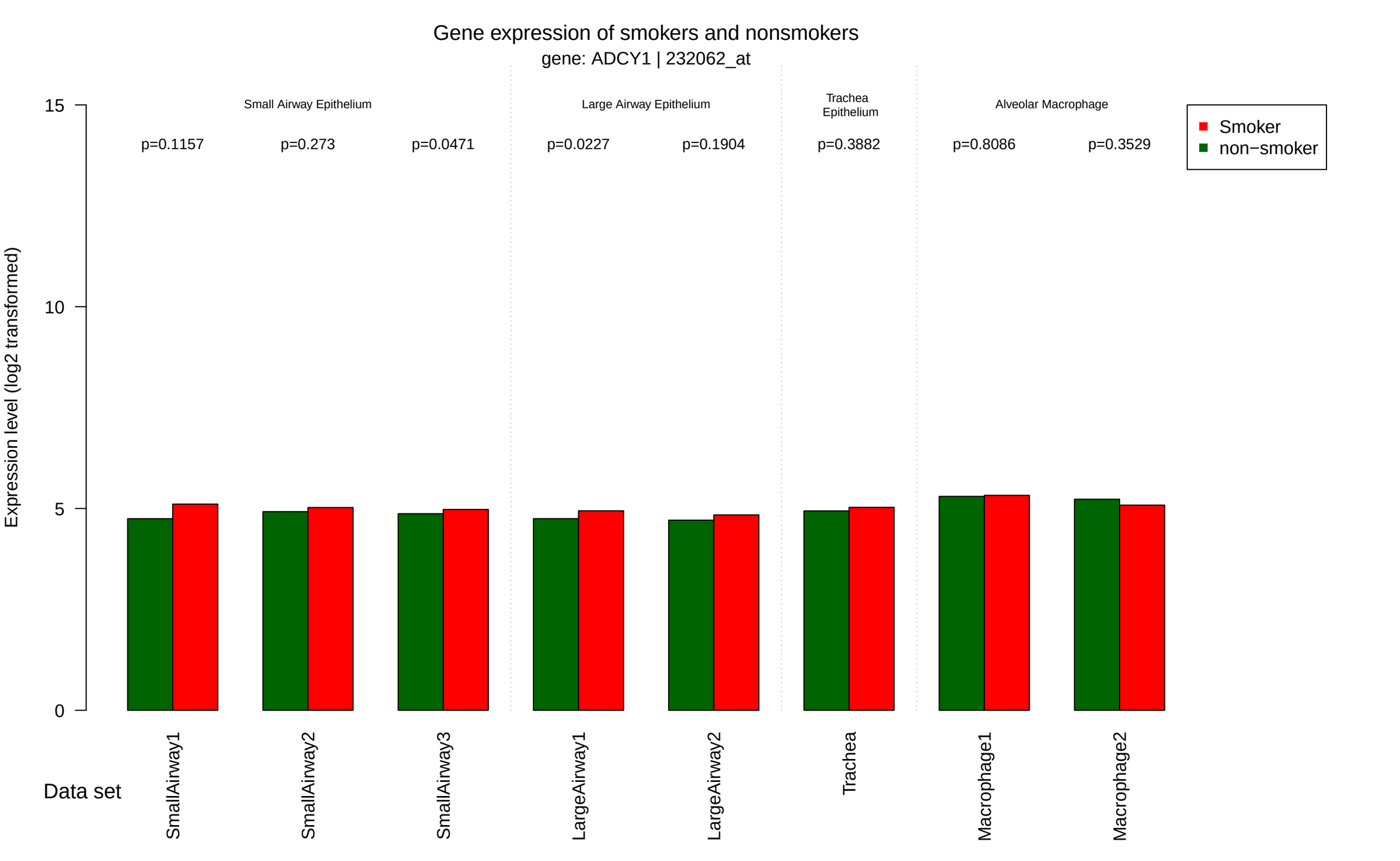
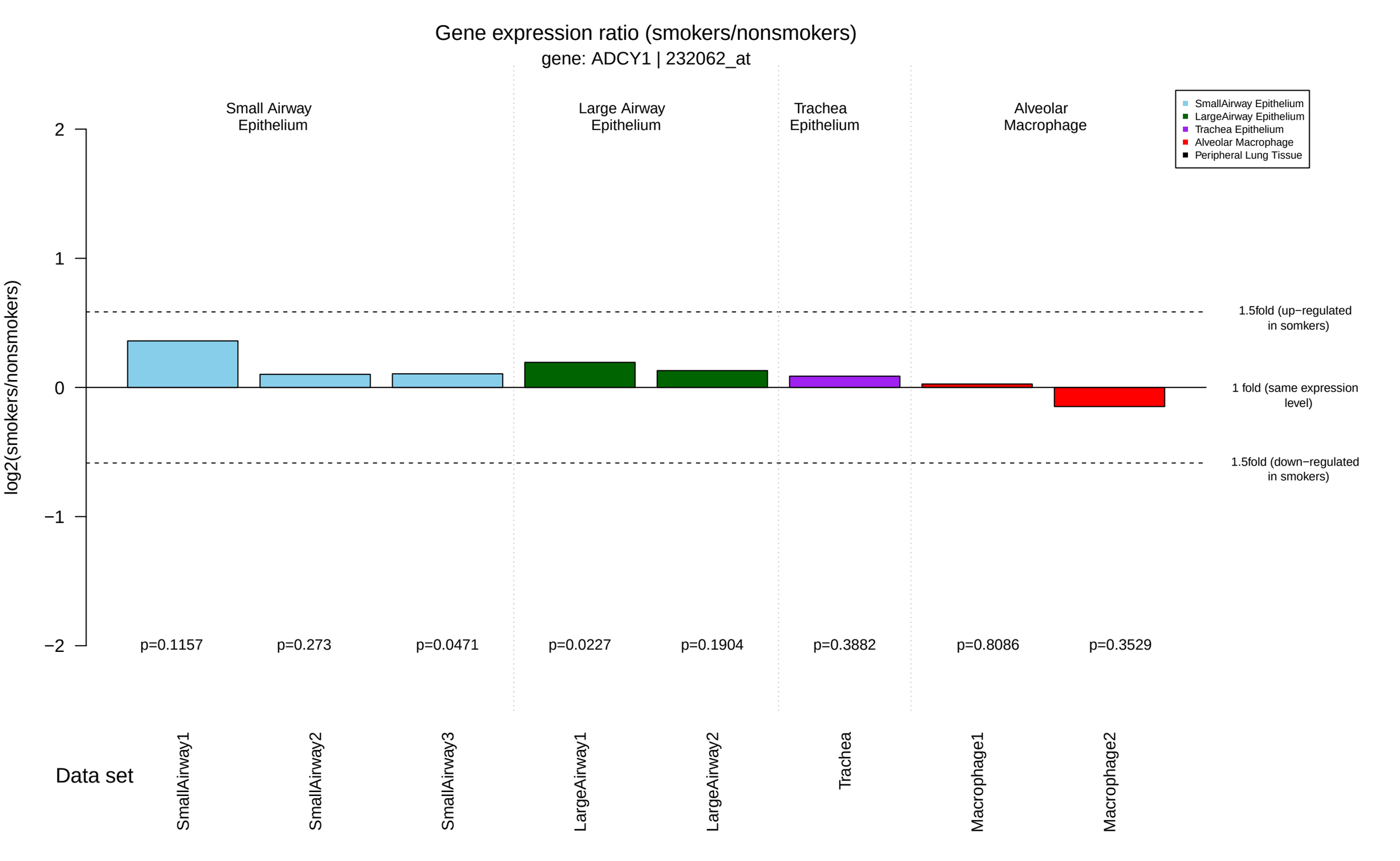
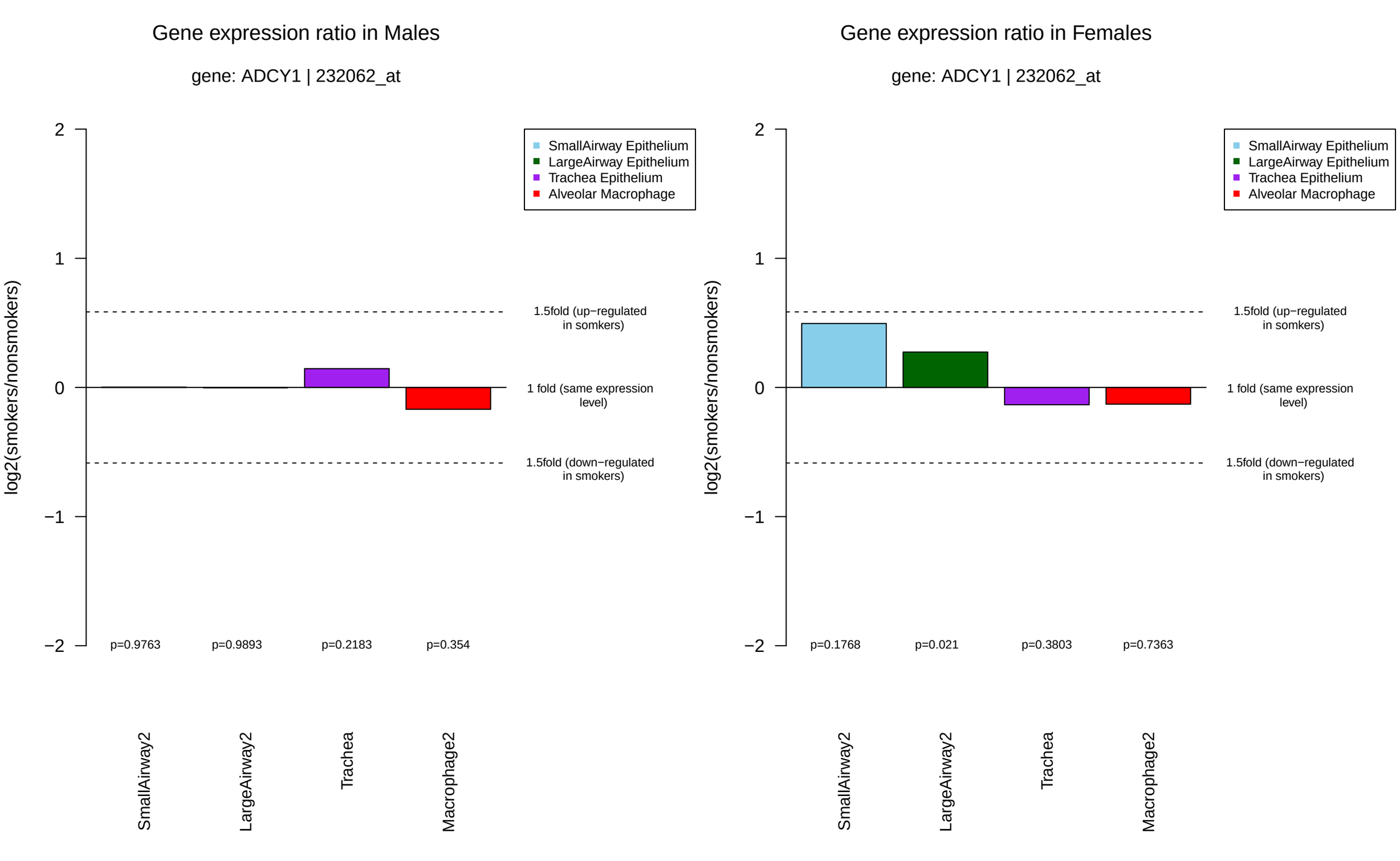
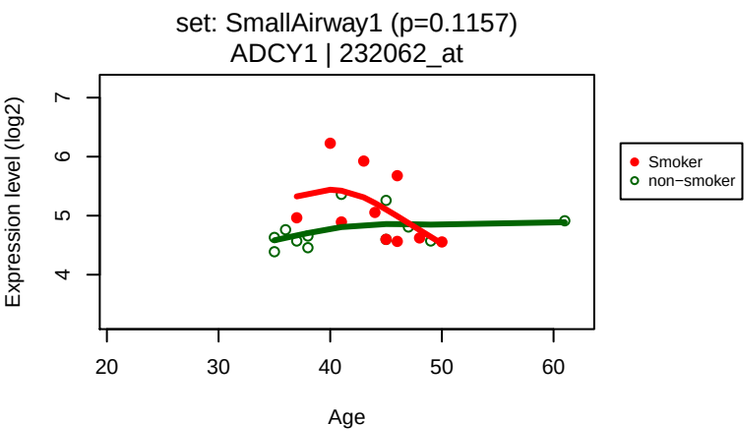
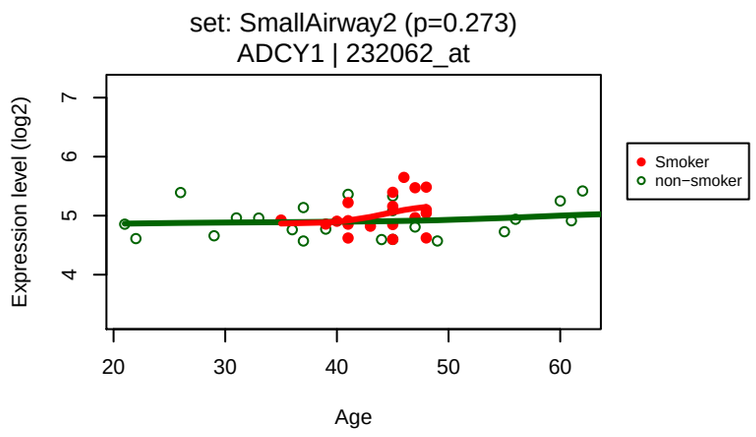
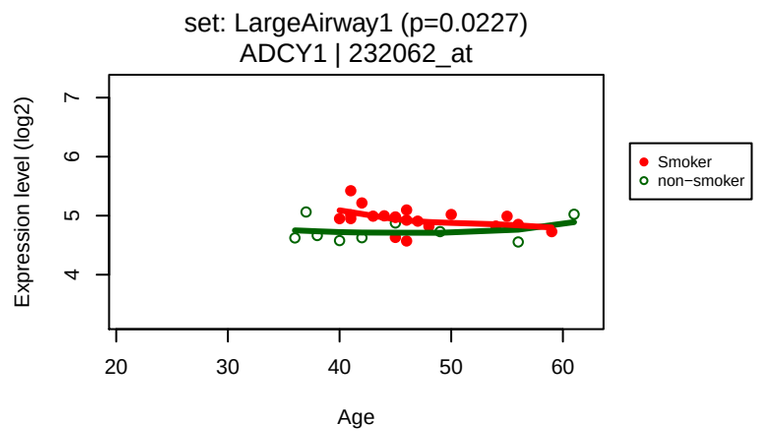

Gene expression of smokers and nonsmokers
gene: ADCY1 | 232062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1157
p=0.273
p=0.0471
p=0.0227
p=0.1904
p=0.3882
p=0.8086
p=0.3529
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

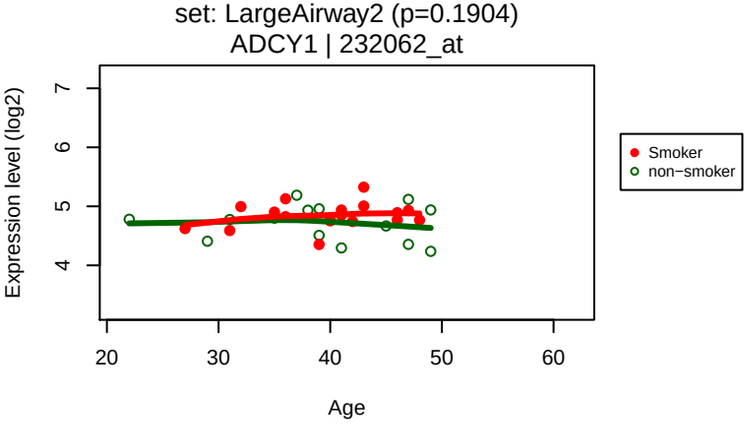

Gene expression ratio (smokers/nonsmokers)
gene: ADCY1 | 232062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1157
p=0.273
p=0.0471
p=0.0227
p=0.1904
p=0.3882
p=0.8086
p=0.3529
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

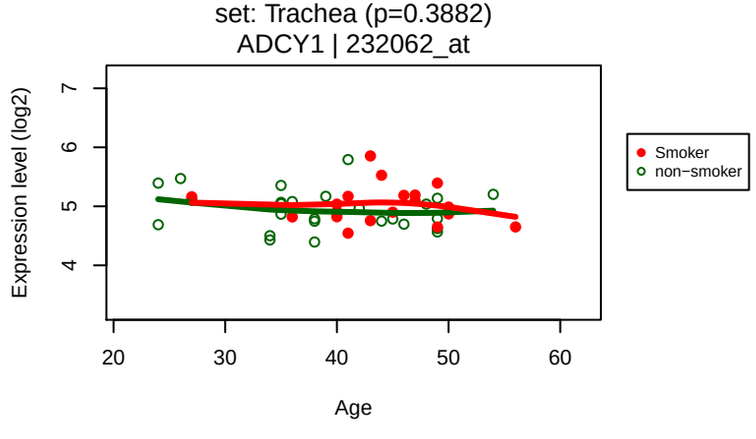

Gene expression ratio in Males
gene: ADCY1 | 232062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9763
p=0.9893
p=0.2183
p=0.354
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY1 | 232062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1768
p=0.021
p=0.3803
p=0.7363
SmallAirway2
LargeAirway2
Trachea
Macrophage2

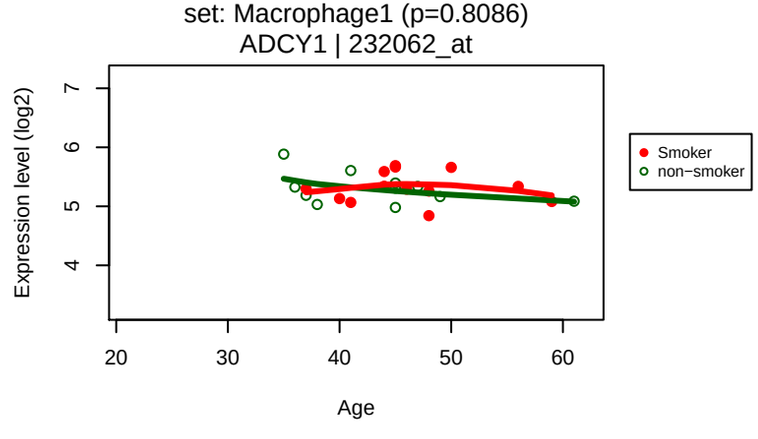

set: SmallAirway1 (p=0.1157)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker

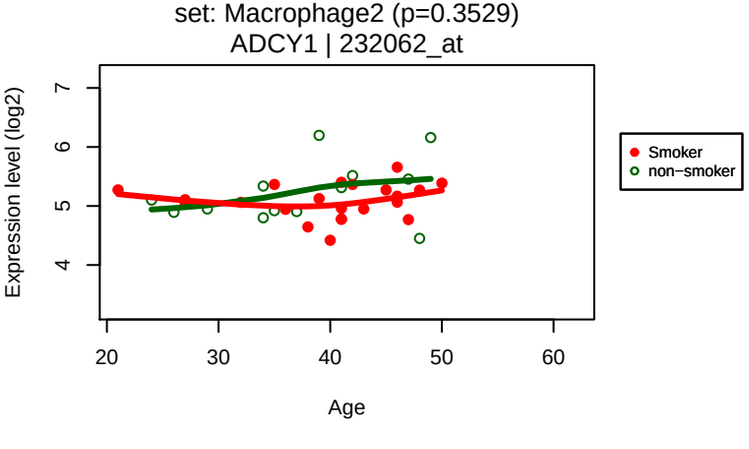

set: SmallAirway2 (p=0.273)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker

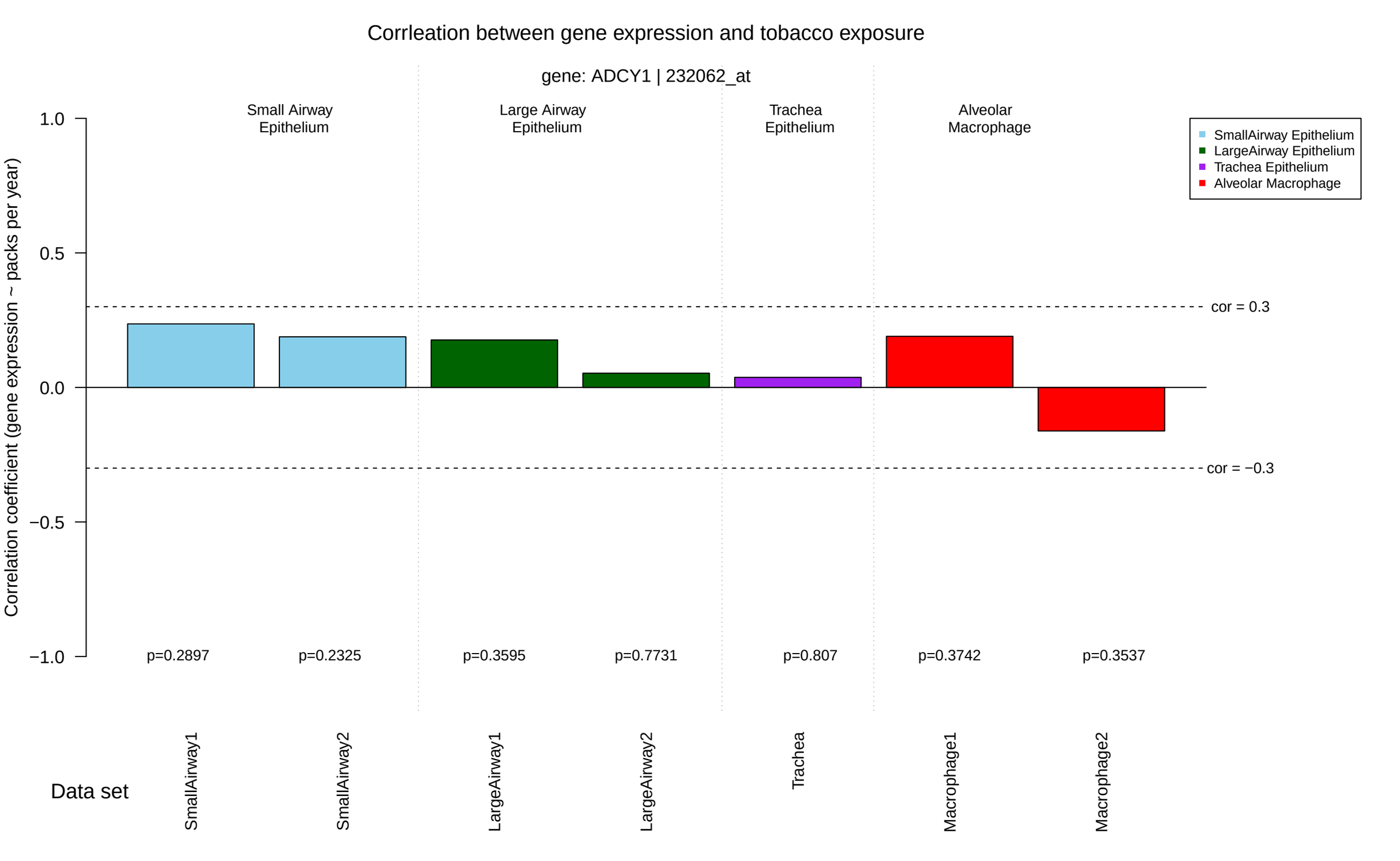

set: LargeAirway1 (p=0.0227)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1904)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3882)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8086)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3529)
ADCY1 | 232062_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY1 | 232062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2897
p=0.2325
p=0.3595
p=0.7731
p=0.807
p=0.3742
p=0.3537
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2